

Priors

Key ideas and concepts

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Outline

- The purpose of priors
- Different kinds of priors and their applications
 - Conjugate priors
 - Jeffreys priors
 - Informative priors
- Hyperpriors and variance priors

A motivating question

Suppose we are studying the effect of air pollution on hospital admissions.

Should we analyse the data as if no previous studies, mechanistic knowledge, or expert understanding existed?

Bayesian inference provides a way to incorporate such information explicitly through the prior.

What is a prior?

A **prior distribution** describes what is known, believed, or considered plausible about an unknown parameter before observing the current data.

It may reflect:

- previous empirical studies,
- expert elicitation,
- biological or physical constraints,
- or weak assumptions used to regularize estimation.

The Bayesian framework

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$$

- $p(\theta)$: prior distribution
- $p(y \mid \theta)$: likelihood
- $p(\theta \mid y)$: posterior distribution

The posterior combines prior information with information from the observed data.

Why priors matter

Priors are important because:

- scientific inference rarely starts from zero knowledge,
- they allow existing evidence to be incorporated,
- they can stabilize estimation in noisy or sparse settings,
- and they make assumptions explicit.

These issues are especially important in environmental health, where effects are often small and uncertainty is substantial.

Common classes of priors

Priors can be chosen for different reasons:

- mathematical convenience,
- formal default rules,
- weak regularisation,
- or incorporation of external knowledge.

Some of the most commonly discussed classes are:

- conjugate priors
- Jeffreys priors
- vague or noninformative priors
- informative priors

Conjugate priors

A prior is **conjugate** if the posterior belongs to the same distributional family as the prior.

Example:

$$Y \mid \theta \sim \text{Binomial}(n, \theta)$$

$$\theta \sim \text{Beta}(\alpha, \beta)$$

Then:

$$\theta \mid Y \sim \text{Beta}(\alpha + Y, \beta + n - Y)$$

Common conjugate priors

Likelihood	Parameter	Prior	Posterior
Binomial	θ	Beta	Beta
Poisson	λ	Gamma	Gamma
Exponential	λ	Gamma	Gamma
Normal mean (σ^2 known)	μ	Normal	Normal
Multinomial	θ	Dirichlet	Dirichlet

Jeffreys priors

Jeffreys priors are defined using the Fisher information:

$$p(\theta) \propto \sqrt{I(\theta)}$$

They are often used as default or “objective” priors because they are invariant under reparameterization.

Why useful?

- principled default construction
- invariant to one-to-one transformations
- theoretically elegant

Jeffreys priors

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Limitations:

- may be improper
- may be hard to interpret in applied settings
- are not always practically noninformative

Informative priors

An **informative prior** incorporates substantive prior knowledge about a parameter.

Sources may include:

- previous studies
- meta analyses
- expert elicitation
- biological or physical constraints

Informative priors

Example:

$$\beta \sim N(0.01, 0.005^2)$$

This may reflect earlier evidence suggesting a small positive association.

Informative priors

Why useful?

- incorporates existing knowledge
- improves precision
- supports cumulative scientific inference

Limitation: results should be checked with sensitivity analysis.

Example: Air pollution health model

Consider a log-linear health model:

$$Y_i \sim \text{Poisson}(\mu_i)$$

$$\log(\mu_i) = \alpha + \beta X_i + \gamma^\top Z_i$$

- Y_i is a health outcome
- X_i is air pollution exposure
- Z_i are confounders
- β is the association between air pollution and health

Example: Air pollution health model

Previous epidemiologic evidence can inform the prior for β :

$$\beta \sim N(\mu_\beta, \sigma_\beta^2)$$

For example:

$$\beta \sim N(0.01, 0.005^2)$$

Interpretation:

- the prior mean reflects the effect suggested by previous studies
- the prior variance reflects uncertainty in that evidence

Example: Air pollution health model

In many studies, exposure is predicted rather than directly observed.

If $\hat{X}_i$ is an estimated exposure, we may write:

$$X_i \sim N(\hat{X}_i, \tau_i^2)$$

and then use X_i in the health model:

$$\log(\mu_i) = \alpha + \beta X_i + \gamma^\top Z_i$$

This allows uncertainty in the exposure model to be carried through to inference on β .

Example: Air pollution health model

Suppose health outcomes are measured repeatedly over time.

A simple longitudinal model is:

$$Y_{it} \sim \text{Poisson}(\mu_{it})$$

$$\log(\mu_{it}) = \alpha + b_i + \beta X_{it} + \gamma^\top Z_{it}$$

with subject-specific random intercept: $b_i \sim N(0, \sigma_b^2)$

and prior on the heterogeneity parameter: $\sigma_b \sim \text{Half-Normal}(0, 1)$

Here, priors help represent between-subject variability and account for correlation among repeated observations.

Example: ozone and all-cause mortality in Chicago

We fit a simple Bayesian Poisson regression model:

$$Y_t \sim \text{Poisson}(\mu_t)$$

$$\log(\mu_t) = \alpha + \beta x_t$$

where:

- Y_t is daily all-cause mortality
- x_t is daily ozone concentration (scaled)
- β is the log-relative risk associated with ozone

Example: ozone and all-cause mortality in Chicago

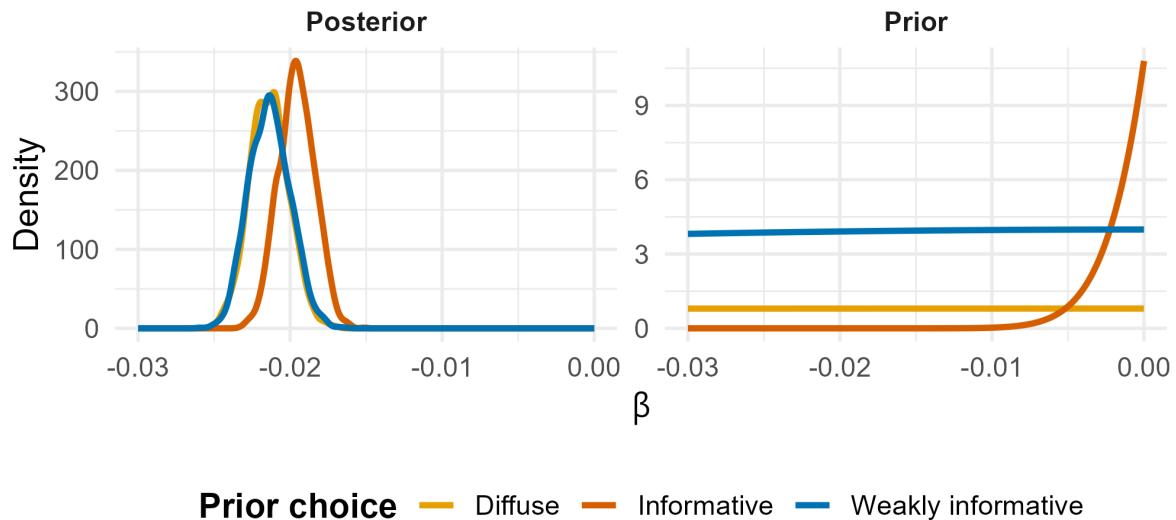
To illustrate the role of the prior, we fit the same model under three different priors for β :

- **Diffuse prior:** $N(0, 0.5)$
- **Weakly informative prior:** $N(0, 0.1)$
- **Informative prior:** $N(0.01, 0.005)$

Prior choice and posterior inference

Priors and posteriors for β

Chicago ozone-mortality example



Example: summer temperature and mortality

In many environmental health settings, the effect of temperature may become stronger only above a critical level.

For summer mortality, a simple scientific question is:

At what temperature does risk begin to increase, and how steep is the increase above that point?

Example: summer temperature and mortality

A Bayesian threshold model allows us to estimate both:

- the **threshold temperature**
- the **slope above the threshold**

A simple linear-threshold model

Let Y_t denote daily all-cause mortality on day t , and let T_t be summer temperature. We consider the model:

$$Y_t \sim \text{Poisson}(\mu_t) \log(\mu_t) = \alpha + \beta(T_t - \psi)_+ (T_t - \psi)_+ = \max(T_t - \psi, 0)$$

- below ψ , temperature has no effect
- above ψ , mortality increases linearly
- β is the slope above the threshold
- ψ is the unknown threshold

Why does the prior on the threshold matter?

The prior matters because:

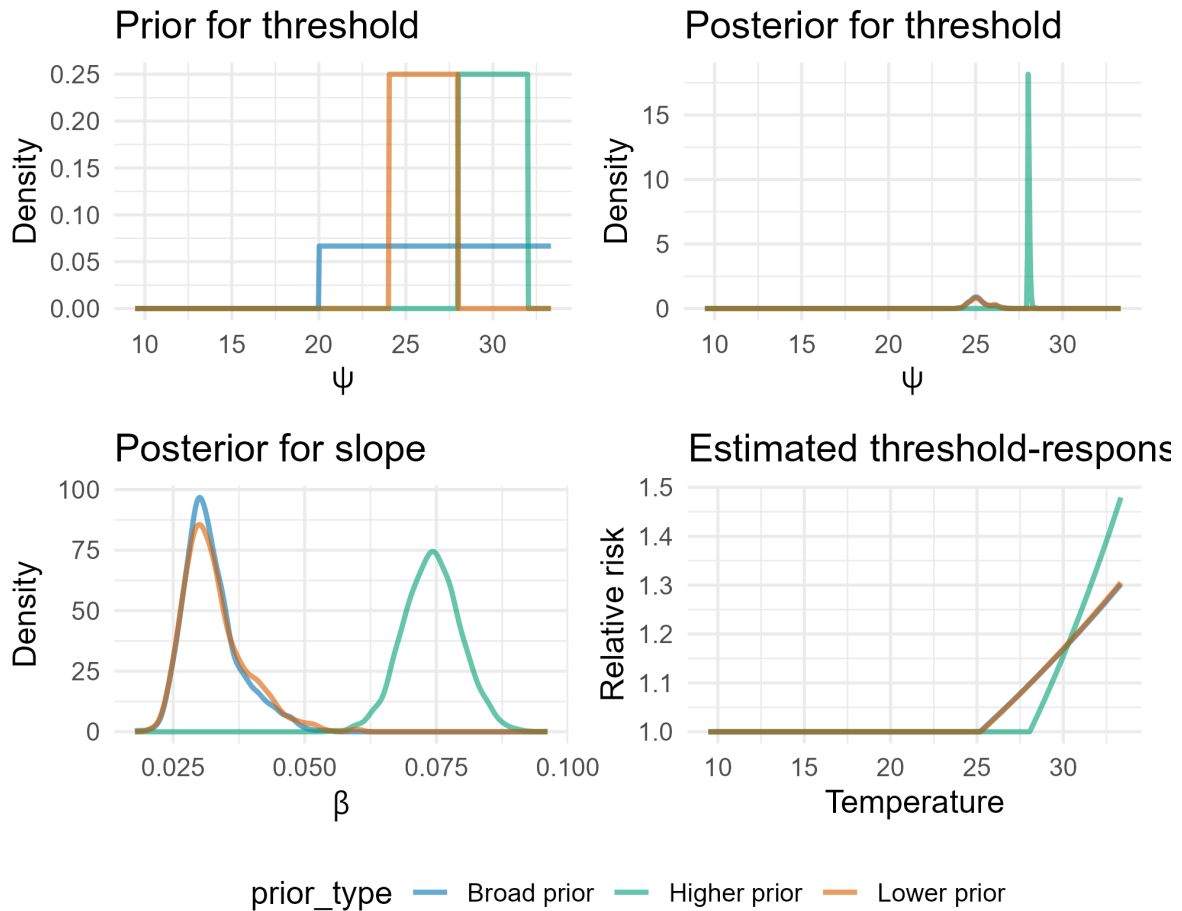
- the threshold may be only weakly identified by the data
- different values of ψ imply different effective exposure-response shapes
- ψ and β are estimated jointly

Why does the prior on the threshold matter?

As a result, the prior on ψ can affect:

- the posterior of the threshold
- the posterior of the slope above threshold

Result



Hyperpriors and variance parameters

In hierarchical Bayesian models, some parameters govern the behavior of other parameters.

These are **hyperparameters**, and priors placed on them are called **hyperpriors**.

Variance hyperparameters are especially important because they control:

- pooling
- smoothness
- model complexity

What is a hyperprior?

For example, in a hierarchical model:

$$b_i \sim N(0, \sigma_b^2)$$

the parameter σ_b determines how much the random effects vary.

If we assign a prior to σ_b , such as

$$\sigma_b \sim \text{Half-Normal}(0, 1)$$

then this prior is a **hyperprior**.

Variance controls pooling

In a hierarchical prior:

$$b_i \sim N(0, \sigma_b^2)$$

- small σ_b implies stronger shrinkage toward zero
- large σ_b allows more heterogeneity across groups

So the hyperprior on σ_b controls the amount of **partial pooling**.

Variance controls smoothness

In a smoothing model, for example

$$f_t \sim N(f_{t-1}, \sigma_f^2)$$

the variance parameter σ_f^2 controls how much the function can change over time.

- small σ_f gives a smoother function
- large σ_f allows a rougher function

A prior on σ_f is therefore a prior on **roughness**.

Hyperpriors in Gaussian processes

For a Gaussian process,

$$f(\cdot) \sim GP(0, K(\cdot, \cdot))$$

the covariance kernel often depends on hyperparameters such as:

- a variance parameter
- a length-scale parameter

Hyperpriors in Gaussian processes

For example:

$$K(x, x') = \sigma^2 \exp\left(-\frac{(x - x')^2}{2\ell^2}\right)$$

Here σ^2 controls variability and ℓ controls smoothness.

Positivity and parameterization

Variance and scale parameters must be positive:

$$\sigma > 0, \quad \sigma^2 > 0$$

Common parameterizations include:

- variance σ^2
- standard deviation σ
- precision $\tau = 1/\sigma^2$

In practice, priors on the standard deviation are often easier to interpret.

Common prior choices for σ

Common priors for scale parameters include:

- half-normal
- half-Student- t
- bounded uniform priors

Examples:

$$\sigma \sim \text{Half-Normal}(0, 1)$$

$$\sigma \sim \text{Half-}t(\nu, 0, s)$$

The old Gamma prior on precision

Historically, one often used:

$$\tau = \frac{1}{\sigma^2} \sim \text{Gamma}(0.001, 0.001)$$

as a supposedly vague prior.

But this can be misleading.

A prior that looks diffuse on the precision scale may be highly informative on the variance scale, leading to undesirable behavior in hierarchical models.

Example: regional mortality in Italy

Suppose we observe mortality counts across Italian regions, together with expected counts based on a reference population.

A natural question is whether mortality risk varies across regions after accounting for the expected number of deaths.

To model this heterogeneity, we introduce a region-specific random intercept.

This provides a simple example for studying how the prior on a variance parameter affects **pooling**.

Hierarchical model

Let Y_i be the observed deaths in region i and E_i the expected deaths.

We model:

$$Y_i \sim \text{Poisson}(\mu_i) \mu_i = E_i \lambda_i \log(\lambda_i) = \alpha + b_i b_i \sim N(0, \sigma_b^2)$$

- λ_i is the region-specific relative risk
- b_i is the region-specific deviation on the log scale
- σ_b controls the heterogeneity across regions

The prior on σ_b controls pooling

The parameter σ_b determines how much the regional effects b_i are allowed to vary.

- small σ_b implies stronger shrinkage toward the overall mean
- large σ_b allows more regional heterogeneity

The prior on σ_b controls pooling

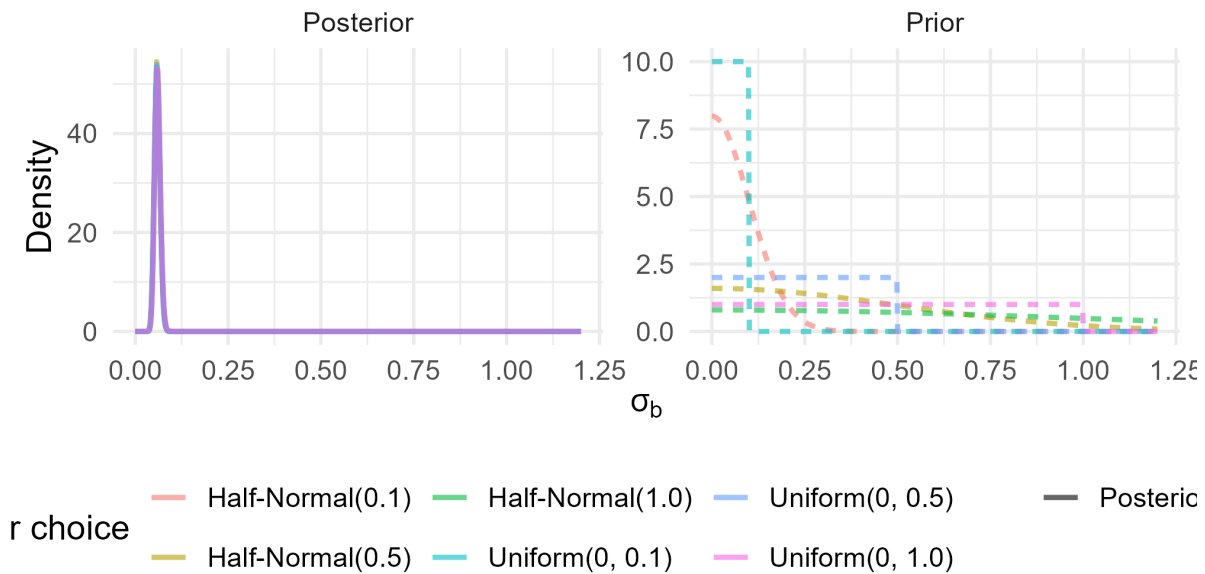
So the prior on σ_b directly influences the amount of **partial pooling**:

- stronger concentration near zero $\rightarrow$ more pooling
- broader support for larger values $\rightarrow$ less pooling

σ_b important in hierarchical models.

Results

Prior and posterior for σ_b



How should we tune the variance prior?

A useful strategy is:

1. think on the scale of σ
2. ask what amount of heterogeneity or roughness is plausible
3. choose a weakly informative prior that respects that scale
4. use prior predictive checks
5. examine sensitivity to alternative priors

Variance priors should be scientifically interpretable, not chosen only for convenience.

What we have seen

- Priors are a central part of Bayesian inference. Different priors serve different purposes:
 - conjugate and default priors
 - informative priors based on existing evidence
 - hyperpriors for hierarchical and smoothing models

- Priors on variance parameters are especially important because they control:
 - pooling
 - smoothness
 - model complexity

Key messages and next step

Good practice

- choose priors on an interpretable scale
- respect parameter constraints, such as positivity
- use prior predictive thinking
- assess sensitivity to alternative reasonable priors :::

Getting ready for the lab

The lab for this session

- This lab will involve taking some models and concepts from the Priors lecture and exploring how different prior choices affect posterior inference in **NIMBLE**.
- During this lab session, we will:
 1. Fit a Bayesian Poisson regression model for ozone and all-cause mortality in Chicago;
 2. Compare posterior inference for the ozone effect under diffuse, weakly informative, and informative priors;
 3. Fit a linear-threshold model for summer temperature and mortality, and see how the prior on the threshold affects both the threshold and the slope above it; and
 4. Use basic MCMC diagnostics to assess convergence and sampling quality.

Questions?